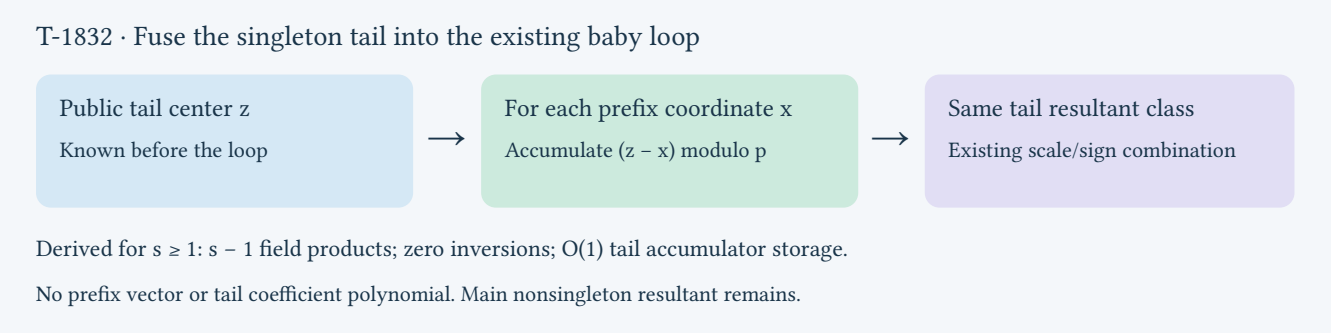
Streamed singleton resultants

T-1832 · 2026-11-07 · Fused public source tail

The public sign source now computes its singleton tail resultant during the existing baby-point loop. It no longer collects a separate prefix-coordinate vector, constructs its coefficient polynomial or traverses that vector again. A general singleton specialization also serves either orientation of the root-resultant API.



Identity and convention. For monic root products f and g, the source uses Res(g,f), the product of point-minus-root differences. With one point y this is the product of y−x over roots; with one root x it is the product of y−x over points. Multiplicities and zero factors are preserved. Empty inputs still return the existing error. Noncanonical u64 inputs are reduced before subtraction.

Derived cost. For N nonempty factors, initialize from the first difference and multiply N−1 remaining factors: exactly N−1 field products, zero inversions, constant auxiliary scalar storage. The API builds no coefficient polynomials on this path. Source fusion applies the same rule to s tail entries, removing the prefix temporary. Input slices and other source/workspace arrays remain charged separately. Nonsingleton main resultants retain the general algorithm and trace-dependent costs.

Exact software check	Count	Coverage
Singleton cases	8,320	Both orientations; lengths 1–65; 8 primes
Zero / orientation controls	3,852 / 2,276	Common roots; wrong sign detectable
General-route / empty controls	128 / 24	Fallback equality; preserved errors
Large-prime boundary control	1	Prime 2^61−1; noncanonical u64 inputs
Source points / windows	1,108 / 24	Direct products, covariance, signs
Additional recoveries	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 calls

Evidence. Every singleton case asserts N−1 products, zero inversions and zero root-polynomial calls. The unchanged general coefficient resultant and direct field product provide controls. The pre-fusion API checks and fused-source checks both passed; earlier source snapshots and receipts are retained. The inherited 160,660 division equalities, 4,100 lifetime plans and eight preparation controls pass. Final normal build pins seven sources with audit cfg disabled.

Scope and custody. Original 4,563-case cohort unchanged. No timing comparison, external benchmark, cloud run or full Cargo rerun. Prime/coprime source and word-size assumptions remain. PCC-20261107-8783a9 is unsigned producer evidence. See DERIVATION.md, validation-02, recovery-02 and normal-build-02 for formulas, commands and input hashes.

Sources and next work. The standard root-resultant identity is documented by FLINT; prior source/accounting code fixes orientation and integration. KB unavailable; no novelty claim. Next inspect the nonsingleton remainder trace and data lifetimes to target larger coefficient intermediates. Historical reports and canonical graphs remain unchanged.